

Srihan Vege

West Lafayette, Indiana

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EDUCATION

Purdue University

Aug 2025 – Dec 2028 (Expected)

B.S. in Computer Science & Mathematics

West Lafayette, Indiana

- **Relevant Coursework:** Linear Algebra, Multivariable Calculus, Differential Equations, Discrete Mathematics, Statistics, C Programming, Java Programming, Programming I
- **Activities:** Hack the Future, Rocket Propulsion Lab

INDUSTRY EXPERIENCE

Microsoft

Aug 2026 – Present

Data Science Researcher, Minecraft

West Lafayette, IN

- Building ML pipelines for content classification and trend analysis on the Minecraft Marketplace through Purdue's Data Mine Corporate Partners program

NuStudio.AI

May 2026 – Sep 2026

AI/ML Intern

Chicago, IL

- Engineered an LLM-powered agent for automated cross-language code migration under the Apex team
 - Reached 95% translation accuracy on test conversions
- Refined agent behavior, tested edge cases, and validated outputs across Cortava AI's credit union AI product suite
 - Improved tool reliability and surfaced failure modes across core platform workflows
- Designed and implemented REST endpoints for Sarvos's Action Broker, covering typed action-proposal contracts and validate/execute flows across billing, document, and matter-intake domains
 - Enabled deterministic, auditable execution of agent-proposed actions across all endpoint domains

Bioqore

Jun 2026 – Aug 2026

Software Engineering Intern

Boston, MA

- Developed a synthetic biomanufacturing data framework in Python using SDV's Gaussian Copula and CTGAN models
 - Benchmarked models on statistical fidelity, domain validity, utility, and privacy to select the optimal generator for upstream (USP) process data
 - Achieved 85% fidelity against real process data
- Designed a biological validity rule engine and model bake-off harness enforcing domain plausibility constraints (titer/IVCD consistency, metabolite endpoint bounds) as hard gates
 - Flagged 93% of implausible synthetic records across 8 validation rules before model selection

Vela (Y Combinator W26)

Jan 2026 – Feb 2026

Analytics Engineering (Contract)

San Francisco, CA

- Built and deployed Google Analytics 4 event tracking infrastructure in TypeScript across Vela's website to measure user workflows, conversion funnels, and retention

RESEARCH EXPERIENCE

AlgoVerse AI Research

Aug 2024 – Apr 2025

Machine Learning Researcher

Palo Alto, CA

- Built a multi-turn LLM evaluation harness to quantify sycophancy-driven accuracy degradation across conversation turns
 - Prompting strategies reduced accuracy degradation from sycophancy by 43%
- Designed and implemented static and rationale-based feedback pipelines to simulate human-like multi-turn interactions
- Joint first author of *TRUTH DECAY (NAACL SRW 2025)*; 25+ citations, including work from Harvard and Google DeepMind

Northwestern University, Bagci Lab

Jun 2024 – Feb 2025

Machine Learning Engineer, Machine & Hybrid Intelligence Lab (MHIL)

Chicago, IL

- Developed *ContrastIQ*, a CT imaging model combining a CNN with pharmacokinetic priors to predict aortic peak contrast enhancement timing for radiologists (PyTorch, MONAI)
 - Predicted vs. actual peak time differed by a mode of 0.3 s, a 97% reduction from previous test results
 - Improved prediction accuracy by 50% on 272 patient CT scans, reducing radiodensity error from 200 HU to 100 HU
- Second author on manuscript submitted to **IEEE EMBC**; patent pending (see Publications)

Stanford University, ML / Molecular Imaging Fellowship

Jun 2023 – Aug 2023

Research Student, advised by Dr. Frezghi Habte

Stanford, CA

- Studied applications of deep learning to PET, MRI, and SPECT molecular imaging; surveyed CNN- and RNN-based imaging tools

PROJECTS

Credit Card Fraud Detection | Python, scikit-learn, pandas, XGBoost

May 2026

- Developed a machine learning pipeline for real-time credit card fraud detection using classification models and anomaly detection techniques on a highly imbalanced dataset of 284,807 transactions
 - Achieved 97.6% ROC AUC and 86% F1-score on the fraud class (88% precision, 84% recall)
 - Developed and compared multiple models, including Logistic Regression and XGBoost, to optimize predictive performance

REALestate.ai | Python, Machine Learning

2024

- Built an AI-powered real estate valuation system that automates property value estimation by combining tabular property features with location and market signals

PUBLICATIONS & PATENTS

Liu, A., Jain, S., Takuri, T., **Vege, S.**, Akalin, A., Zhu, J., O'Brien, S., Sharma, V. (2025). *TRUTH DECAY: Quantifying Multiturn Sycophancy in Language Models*. **NAACL 2025 Student Research Workshop**. arXiv:2503.11656. Joint first author.

ContrastIQ: CNN-based prediction of aortic peak contrast enhancement timing in CT imaging. Second author. Manuscript submitted to **IEEE EMBC**; **patent pending**. Northwestern University, Bagci Lab.

LEADERSHIP & SERVICE

FireFly EDU (501(c)(3) Nonprofit)

Jun 2023 – Aug 2024

Co-founder & Vice President

Remote

- Co-founded and led an education nonprofit serving underserved youth; recruited and managed an executive team across marketing and finance
- Taught computer science, mathematics, biology, English, and physics to students aged 5 to 15 at Heal Paradise School

TECHNICAL SKILLS

Languages: Python, TypeScript, JavaScript, Java, C, SQL

Machine Learning: PyTorch, scikit-learn, XGBoost, MONAI, pandas, NumPy, LLM agents & agentic workflows, LLM evaluation, synthetic data (SDV, CTGAN, Gaussian Copula)

Frontend: React, Next.js, Tailwind, HTML, CSS, Browser APIs

Backend & Tools: Node.js, REST API design, MongoDB, Git, Linux, Google Analytics 4